

U.S. Futures and Options Package

Contents:

Cheapest to Deliver Summary Report
Basis Reference Sheet
Treasury Futures Roll Report
Treasury Futures CTD Basis Convergence Report
Treasury Futures Butterfly Report
Cross Market Futures Matrix
Cross Market Volatility Spreads Report
Short-Dated SOFR Swaption Volatility Report
Volatility Summary Report
Option Market Summary Report

SOFR Futures Relative Value Report
Swap Spread Curve Report
SOFR Risk-adjusted Carry Report
Money Market Futures Calendar Spread Report
SOFR Future Pack
SOFR Future Butterflies Report
Treasury Yield Curve Report
Maturity Structure of Treasury Volatility
Treasury OTC and Exchange Volatility Report

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Derivatives Strategy

	Treasury Bond					Treasury Note				5 Year Treasury Note			
	Jan25	Feb25	Mar25	Apr25	Jun25	Jan25	Feb25	Mar25	Apr25	Jan25	Feb25	Mar25	Apr25
VOLATILITY Implied (%)													
Current	8.39	9.88	10.46	11.00	11.23	4.56	5.41	5.77	5.99	3.03	3.69	3.89	4.02
Change (1d)	-0.62	-0.29	-0.15	-0.18	-0.18	-0.52	-0.24	-0.19	-0.17	-0.52	-0.22	-0.17	-0.16
20 day High	11.72	11.91	12.20	12.42	12.47	6.50	6.41	6.55	6.76	4.43	4.41	4.49	4.55
20 day Low	8.39	9.88	10.46	11.00	11.23	4.56	5.41	5.77	5.99	3.03	3.69	3.89	4.02
Business Day	8.39	10.09	10.61	11.06	11.23	4.56	5.53	5.86	6.02	3.03	3.77	3.95	4.04
Historical (%)													
10 day	8.09	8.09	8.09	8.09	8.09	3.70	3.70	3.70	-	2.45	2.45	2.45	-
20 day	9.30	9.30	9.30	9.56	9.56	4.12	4.12	4.12	-	2.57	2.57	2.57	-
50 day	10.50	10.50	10.50	10.49	10.49	5.32	5.32	5.32	-	3.40	3.40	3.40	-
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)													
3 Year High	21.64	17.91	16.94	16.27	85.14	15.26	11.22	10.01	9.32	12.64	9.26	8.21	7.53
3 Year Average	12.31	12.53	12.60	12.64	16.97	6.97	7.13	7.19	7.21	4.90	5.03	5.09	5.11
3 Year Low	4.98	7.50	8.41	9.16	9.38	2.46	3.82	4.60	4.81	1.73	2.74	3.19	3.07
% Historic Above Implied	89.86	85.11	85.64	76.15	77.08	89.33	83.53	83.14	80.11	88.67	81.16	82.87	84.58
BASIS POINT VOLATILITY Implied (bp) per day													
Current	4.5	5.3	5.7	5.7	5.9	4.8	5.8	6.2	6.1	5.0	6.1	6.4	6.5
Change (1d)	-0.4	-0.2	-0.1	-0.2	-0.2	-0.6	-0.3	-0.2	-0.2	-0.9	-0.4	-0.3	-0.3
Change(5d)	-1.0	-0.5	-0.3	-0.4	-0.4	-1.6	-0.9	-0.7	-0.6	-2.1	-1.0	-0.7	-0.7
Change(20d)	-1.7	-1.0	-0.8	-0.6	-0.5	-1.9	-1.0	-0.6	-0.6	-2.2	-1.1	-0.9	-0.7
20 day High	6.2	6.3	6.5	6.4	6.5	6.9	6.8	7.0	6.8	7.3	7.2	7.3	7.4
20 day Low	4.5	5.3	5.7	5.7	5.9	4.8	5.8	6.2	6.1	5.0	6.1	6.4	6.5
Historical (bp) per day													
10 day	4.6	4.6	4.6	4.6	4.6	3.9	3.9	3.9	3.9	3.9	3.9	3.9	3.9
20 day	5.3	5.3	5.3	5.3	5.3	4.2	4.2	4.2	4.2	4.0	4.0	4.0	4.0
50 day	5.9	5.9	5.9	5.9	5.9	5.7	5.7	5.7	5.7	5.4	5.4	5.4	5.4
Volatility Cones (bp) per day													
3 Year High	13.0	11.1	10.2	9.8	8.7	16.3	12.0	10.6	10.1	19.7	14.3	12.7	11.6
3 Year Average	6.9	7.0	7.1	7.2	7.3	7.4	7.5	7.5	7.5	7.6	7.8	7.9	7.9
3 Year Low	2.7	4.2	4.6	5.0	5.0	3.2	4.3	4.9	4.9	2.9	4.0	4.8	4.5
% Historic Above Implied	91.0	88.1	87.1	88.8	88.5	88.1	82.9	81.8	85.2	84.5	76.5	77.6	78.8
Yield Change Required to... (bp)													
Offset 1 C-day Decay	3.5	4.2	4.5	4.5	4.8	3.8	4.5	4.9	4.6	3.8	4.7	5.1	5.0
Offset 1 B-day Decay	4.3	5.1	5.4	5.5	5.8	4.5	5.4	5.9	5.5	4.6	5.7	6.1	6.0
Offset 1 Month Decay	-	21.2	24.1	24.7	26.5	-	22.0	25.6	25.1	-	23.4	26.7	27.1
Break Even at Expiry	11.4	22.4	30.5	36.2	48.6	11.9	23.9	33.0	36.4	12.3	25.2	34.3	39.8
BPV per Contract	\$146.06	\$146.06	\$146.06	\$156.23	\$156.23	\$70.86	\$70.86	\$70.86	\$78.21	\$44.30	\$44.30	\$44.30	\$47.47
Cal/Bus Days to Expiry	16/11	44/29	72/48	100/68	163/112	16/11	44/29	72/48	100/68	16/11	44/29	72/48	100/68
Futures Price	118-05	118-05	118-05	118-08	118-08	110-22	110-22	110-22	110-26+	107-10+	107-10+	107-10+	107-16+

VOLATILITY is price volatility for bond, note and index options; yield volatility for money market options

Volatility Cones are based on number of days to expiration. For example, if 40 days remain, we look at actual volatility of the nearby contract in every 40 day period over the last 2 years

% Historical Above Implied shows how often actual volatility exceeded current implied volatility. A higher number means cheaper options.

BASIS POINT VOLATILITY is basis point yield volatility for all options.

Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options

* Historical volatility is taken from old otr bond This

Derivatives Strategy

	Secured ON Financing Rate 3M													
	Dec24	Jan25	Feb25	Mar25	Apr25	May25	Jun25	Sep25	Dec25	Mar26	Jun26	Sep26	Dec26	Mar27
VOLATILITY Implied (%)														
Current	7.74	13.83	14.14	13.68	20.79	20.59	20.37	24.35	27.55	28.74	29.38	29.86	30.42	30.56
Change (1d)	-6.43	-2.90	-2.03	-1.72	-1.87	-1.66	-1.77	-1.32	-0.91	-1.03	-1.00	-1.00	-0.76	-0.42
20 day High	16.66	19.09	18.82	18.62	24.92	24.30	24.12	27.17	29.71	31.04	31.55	31.73	31.72	31.84
20 day Low	7.74	13.83	14.14	13.68	20.79	20.59	20.37	24.35	27.47	28.74	29.38	29.86	30.09	29.98
Business Day	6.42	14.04	14.25	13.78	20.81	20.61	20.40	24.41	27.53	28.79	29.36	29.86	30.38	30.56
Historical (%)														
10 day	7.02	8.00	8.94	12.61	12.94	13.44	15.45	17.57	18.26	18.74	18.74	19.57	19.42	18.96
20 day	8.56	8.97	10.02	13.30	13.58	9.68	16.69	18.15	18.42	18.65	18.83	19.03	19.14	19.01
50 day	12.45	14.95	17.85	21.96	11.82	6.03	27.01	28.62	28.61	28.03	27.58	27.15	25.71	24.66
2Yr Event Risk	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Volatility Cones (%)														
3 Year High	306.06	257.65	178.28	154.05	149.93	141.38	132.83	117.85	97.93	81.75	68.49	59.30	55.79	53.51
3 Year Average	32.36	44.32	45.90	46.79	47.94	48.15	48.48	47.51	46.36	45.19	44.16	43.47	42.90	42.82
3 Year Low	0.00	11.99	14.01	19.15	26.28	26.20	30.08	32.64	33.84	33.92	33.83	34.28	34.57	34.96
% Historic Above Implied	80.37	97.89	99.87	100.00	100.00	97.36	94.86	86.69	78.26	70.22	61.79	53.62	45.19	37.15
BASIS POINT VOLATILITY Implied (bp) per day														
Current	2.1	3.6	3.7	3.6	5.2	5.1	5.1	6.0	6.6	6.9	7.0	7.2	7.3	7.4
Change (1d)	-1.8	-0.8	-0.5	-0.5	-0.5	-0.4	-0.4	-0.3	-0.2	-0.2	-0.2	-0.1	0.0	0.1
Change(5d)	-1.5	-1.4	-1.1	-1.0	-1.0	-0.8	-0.8	-0.6	-0.4	-0.3	-0.3	-0.2	-0.1	-0.1
Change(20d)	-0.4	-1.3	-1.2	-1.2	-1.0	-	-0.9	-0.8	-0.6	-0.5	-0.5	-0.4	-0.2	-0.1
20 day High	4.7	5.0	5.0	4.9	6.3	6.2	6.1	6.7	7.1	7.4	7.5	7.5	7.6	7.6
20 day Low	2.1	3.6	3.7	3.6	5.2	5.1	5.1	6.0	6.6	6.9	7.0	7.2	7.3	7.3
Historical (bp) per day														
10 day	2.0	3.3	3.3	3.3	3.9	3.9	3.9	4.3	4.4	4.4	4.4	4.5	4.5	4.4
20 day	2.4	3.5	3.5	3.5	4.2	4.2	4.2	4.5	4.5	4.5	4.5	4.5	4.5	4.5
50 day	3.3	5.4	5.4	5.4	6.2	6.2	6.2	6.3	6.2	6.0	5.9	5.8	5.6	5.4
Volatility Cones (bp) per day														
3 Year High	31.5	14.5	12.7	12.0	10.5	10.6	10.3	9.7	9.3	9.2	8.9	8.7	10.2	10.0
3 Year Average	6.7	8.7	8.9	8.9	8.9	8.9	8.9	8.8	8.7	8.5	8.4	8.3	8.5	8.9
3 Year Low	0.0	0.2	0.3	0.4	0.4	0.3	0.5	1.9	3.8	5.7	6.4	7.2	7.4	7.5
% Historic Above Implied	75.4	97.6	99.5	99.6	99.5	96.3	93.8	85.6	77.2	69.2	60.7	53.6	45.2	37.2
Yield Change Required to... (bp)														
Offset 1 C-day Decay	1.3	3.0	3.0	2.9	4.2	4.2	4.1	4.8	5.3	5.4	5.5	5.5	5.6	5.6
Offset 1 B-day Decay	1.5	3.6	3.6	3.5	5.1	5.1	5.0	5.8	6.4	6.5	6.6	6.7	6.8	6.7
Offset 1 Month Decay	-	-	15.9	15.7	23.2	23.0	22.8	26.4	29.3	30.1	30.4	30.7	31.1	31.0
Break Even at Expiry	2.0	13.0	19.5	22.5	36.9	41.6	44.5	63.3	79.9	92.5	102.3	111.1	120.0	126.9
BPV per Contract	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00	\$25.00
Cal/Bus Days to Expiry	2/2	30/20	65/44	93/63	121/83	156/107	184/126	275/189	366/251	457/312	548/375	639/438	730/500	821/561
Futures Price	95-20+	95-28	95-28	95-28	96-01+	96-01+	96-01+	96-04+	96-06+	96-08	96-08+	96-09+	96-09	96-09+

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Yield Change Required to... is based on an at-the-money straddle. Yield change for bond, note and money market options; Price change for equity and commodity options

* Historical volatility is taken from old otr bond This